

# Evaluation results (mIoU, OA & per-class IoU)

Parsed from `eval.log` files (and loveda `train.log`, which has no eval run) under `experiments/`. Re-run `scripts/extract_evaluation_tables.py` to refresh.

## Dataset setup (resolution & split sizes)

All images are resized to **1008×1008** (config `dataset.image_size`) for training and evaluation.

| Dataset                 | Resolution (crop) | Train | Val   | Test  |
|-------------------------|-------------------|-------|-------|-------|
| massachusetts_buildings | 1008×1008         | 1,233 | 36    | 90    |
| massachusetts_roads     | 1008×1008         | 9,972 | 126   | 441   |
| potsdam                 | 1008×1008         | 2,765 | 691   | 2,016 |
| uavid                   | 1008×1008         | 8,000 | 2,800 | 150   |
| vaihingen               | 1008×1008         | 276   | 68    | 398   |
| loveda                  | 1008×1008         | 2,522 | 1,669 | —     |
| whu_building            | 1008×1008         | 5,732 | 1,228 | 1,228 |

## Trainable parameters (by method)

| Method         | Trainable params | Total params | Trainable % |
|----------------|------------------|--------------|-------------|
| Linear Probing | 2,299,652        | 456,339,460  | 0.50%       |
| LoRA           | 4,539,139        | 460,877,828  | 0.98%       |
| Adapter        | 337,891          | 456,676,580  | 0.07%       |

(From Massachusetts Buildings runs; slight variation with `num_classes`.)

Summary table (all runs)

| Dataset                 | Method         | mIoU   | OA     |
|-------------------------|----------------|--------|--------|
| loveda                  | Linear Probing | 0.4237 | 0.5838 |
| loveda                  | LoRA           | 0.5663 | 0.7411 |
| loveda                  | Adapter        | 0.5518 | 0.7288 |
| massachusetts_buildings | Linear Probing | 0.7381 | 0.9080 |
| massachusetts_buildings | LoRA           | 0.8278 | 0.9415 |
| massachusetts_buildings | Adapter        | 0.8087 | 0.9335 |
| massachusetts_roads     | Linear Probing | 0.7048 | 0.9638 |
| massachusetts_roads     | LoRA           | 0.8128 | 0.9798 |
| massachusetts_roads     | Adapter        | 0.8077 | 0.9790 |
| potsdam                 | Linear Probing | 0.7186 | 0.8170 |
| potsdam                 | LoRA           | 0.8806 | 0.9306 |
| potsdam                 | Adapter        | 0.8735 | 0.9293 |
| uavid                   | Linear Probing | 0.4554 | 0.7440 |
| uavid                   | LoRA           | 0.6106 | 0.8697 |
| uavid                   | Adapter        | 0.6619 | 0.8932 |
| vaihingen               | Linear Probing | 0.6917 | 0.8186 |
| vaihingen               | LoRA           | 0.8326 | 0.9101 |
| vaihingen               | Adapter        | 0.8152 | 0.9048 |
| whu_building            | Linear Probing | 0.8877 | 0.9682 |
| whu_building            | LoRA           | 0.9492 | 0.9862 |
| whu_building            | Adapter        | 0.9403 | 0.9837 |

Comparison by dataset (methods as columns)

| Dataset                 | Linear Probing (mIoU) | Linear Probing (OA) | LoRA (mIoU) | LoRA (OA) | Adapter (mIoU) | Adapter (OA) |
|-------------------------|-----------------------|---------------------|-------------|-----------|----------------|--------------|
| loveda                  | 0.4237                | 0.5838              | 0.5663      | 0.7411    | 0.5518         | 0.7288       |
| massachusetts_buildings | 0.7381                | 0.9080              | 0.8278      | 0.9415    | 0.8087         | 0.9335       |
| massachusetts_roads     | 0.7048                | 0.9638              | 0.8128      | 0.9798    | 0.8077         | 0.9790       |
| potsdam                 | 0.7186                | 0.8170              | 0.8806      | 0.9306    | 0.8735         | 0.9293       |
| uavid                   | 0.4554                | 0.7440              | 0.6106      | 0.8697    | 0.6619         | 0.8932       |
| vaihingen               | 0.6917                | 0.8186              | 0.8326      | 0.9101    | 0.8152         | 0.9048       |
| whu_building            | 0.8877                | 0.9682              | 0.9492      | 0.9862    | 0.9403         | 0.9837       |

## Per-class IoU by dataset

### loveda

| Class       | Linear Probing | LoRA          | Adapter       |
|-------------|----------------|---------------|---------------|
| background  | 0.3271         | 0.5533        | 0.5473        |
| building    | 0.5934         | 0.6925        | 0.6617        |
| road        | 0.5107         | 0.5874        | 0.5869        |
| water       | 0.5649         | 0.7087        | 0.7054        |
| barren      | 0.1676         | 0.3656        | 0.3472        |
| forest      | 0.3282         | 0.4126        | 0.3984        |
| agriculture | 0.4737         | 0.6441        | 0.6158        |
| <b>mIoU</b> | <b>0.4237</b>  | <b>0.5663</b> | <b>0.5518</b> |
| OA          | 0.5838         | 0.7411        | 0.7288        |

### massachusetts\_buildings

| Class       | Linear Probing | LoRA          | Adapter       |
|-------------|----------------|---------------|---------------|
| background  | 0.8946         | 0.9308        | 0.9216        |
| building    | 0.5817         | 0.7249        | 0.6958        |
| <b>mIoU</b> | <b>0.7381</b>  | <b>0.8278</b> | <b>0.8087</b> |
| OA          | 0.9080         | 0.9415        | 0.9335        |

### massachusetts\_roads

| Class       | Linear Probing | LoRA          | Adapter       |
|-------------|----------------|---------------|---------------|
| background  | 0.9627         | 0.9790        | 0.9782        |
| road        | 0.4469         | 0.6465        | 0.6372        |
| <b>mIoU</b> | <b>0.7048</b>  | <b>0.8128</b> | <b>0.8077</b> |
| OA          | 0.9638         | 0.9798        | 0.9790        |

### potsdam

| Class              | Linear Probing | LoRA          | Adapter       |
|--------------------|----------------|---------------|---------------|
| impervious surface | 0.7268         | 0.9070        | 0.9051        |
| building           | 0.7778         | 0.9612        | 0.9567        |
| low vegetation     | 0.6002         | 0.8078        | 0.8056        |
| tree               | 0.6542         | 0.8089        | 0.7985        |
| car                | 0.8338         | 0.9182        | 0.9016        |
| <b>mIoU</b>        | <b>0.7186</b>  | <b>0.8806</b> | <b>0.8735</b> |
| OA                 | 0.8170         | 0.9306        | 0.9293        |

## uavid

| Class          | Linear Probing | LoRA          | Adapter       |
|----------------|----------------|---------------|---------------|
| building       | 0.6686         | 0.8962        | 0.9191        |
| road           | 0.6596         | 0.8176        | 0.8391        |
| static car     | 0.5989         | 0.7652        | 0.7915        |
| tree           | 0.4111         | 0.6231        | 0.6858        |
| low vegetation | 0.3233         | 0.5047        | 0.5230        |
| human          | 0.4027         | 0.5395        | 0.7027        |
| moving car     | 0.1232         | 0.1276        | 0.1722        |
| <b>mIoU</b>    | <b>0.4554</b>  | <b>0.6106</b> | <b>0.6619</b> |
| OA             | 0.7440         | 0.8697        | 0.8932        |

## vaihingen

| Class              | Linear Probing | LoRA          | Adapter       |
|--------------------|----------------|---------------|---------------|
| impervious surface | 0.7215         | 0.8784        | 0.8721        |
| building           | 0.8128         | 0.9322        | 0.9143        |
| low vegetation     | 0.4987         | 0.7427        | 0.7323        |
| tree               | 0.7262         | 0.8192        | 0.8157        |
| car                | 0.6991         | 0.7905        | 0.7415        |
| <b>mIoU</b>        | <b>0.6917</b>  | <b>0.8326</b> | <b>0.8152</b> |
| OA                 | 0.8186         | 0.9101        | 0.9048        |

## whu\_building

| Class       | Linear Probing | LoRA          | Adapter       |
|-------------|----------------|---------------|---------------|
| background  | 0.9632         | 0.9838        | 0.9809        |
| building    | 0.8123         | 0.9146        | 0.8997        |
| <b>mIoU</b> | <b>0.8877</b>  | <b>0.9492</b> | <b>0.9403</b> |
| OA          | 0.9682         | 0.9862        | 0.9837        |

## Parsed log files

- experiments/sam3\_adapter\_loveda/train.log — loveda / Adapter: mIoU=0.5518, OA=0.7288
- experiments/sam3\_linear\_probing\_loveda/train.log — loveda / Linear Probing: mIoU=0.4237, OA=0.5838
- experiments/sam3\_lora\_loveda/train.log — loveda / LoRA: mIoU=0.5663, OA=0.7411
- experiments/sam3\_adapter\_massachusetts\_buildings/sam3\_adapter\_massachusetts\_buildings\_eval\_2026-03-24\_10-40-26/eval.log — massachusetts\_buildings / Adapter: mIoU=0.8087, OA=0.9335
- experiments/sam3\_linear\_probing\_massachusetts\_buildings/sam3\_linear\_probing\_massachusetts\_buildings\_eval\_2026-03-09\_06-55-50/eval.log — massachusetts\_buildings / Linear Probing: mIoU=0.7381, OA=0.9080

- experiments/sam3\_lora\_massachusetts\_buildings/  
sam3\_lora\_massachusetts\_buildings\_eval\_2026-03-07\_21-54-09/eval.log —  
massachusetts\_buildings / LoRA: mIoU=0.8278, OA=0.9415
- experiments/sam3\_adapter\_massachusetts\_roads/  
sam3\_adapter\_massachusetts\_roads\_eval\_2026-04-08\_19-54-55/eval.log — massachusetts\_roads /  
Adapter: mIoU=0.8077, OA=0.9790
- experiments/sam3\_linear\_probing\_massachusetts\_roads/  
sam3\_linear\_probing\_massachusetts\_roads\_eval\_2026-03-08\_21-48-20/eval.log —  
massachusetts\_roads / Linear Probing: mIoU=0.7048, OA=0.9638
- experiments/sam3\_lora\_massachusetts\_roads/  
sam3\_lora\_massachusetts\_roads\_eval\_2026-03-07\_22-03-41/eval.log — massachusetts\_roads /  
LoRA: mIoU=0.8128, OA=0.9798
- experiments/sam3\_adapter\_potsdam/sam3\_adapter\_potsdam\_eval\_2026-03-16\_23-38-43/eval.log —  
potsdam / Adapter: mIoU=0.8735, OA=0.9293
- experiments/sam3\_linear\_probing\_potsdam/  
sam3\_linear\_probing\_potsdam\_eval\_2026-02-25\_22-40-38/eval.log — potsdam / Linear Probing:  
mIoU=0.7186, OA=0.8170
- experiments/sam3\_lora\_potsdam/sam3\_lora\_potsdam\_eval\_2026-02-25\_15-46-31/eval.log —  
potsdam / LoRA: mIoU=0.8806, OA=0.9306
- experiments/sam3\_adapter\_uavid/eval\_2026-03-24\_10-55-55/eval.log — uavid / Adapter:  
mIoU=0.6619, OA=0.8932
- experiments/sam3\_linear\_probing\_uavid/sam3\_linear\_probing\_eval\_2026-03-15\_17-02-00/  
eval.log — uavid / Linear Probing: mIoU=0.4554, OA=0.7440
- experiments/sam3\_lora\_uavid/sam3\_lora\_uavid\_eval\_2026-03-14\_23-00-44/eval.log — uavid /  
LoRA: mIoU=0.6106, OA=0.8697
- experiments/sam3\_adapter\_vaihingen/sam3\_adapter\_vaihingen\_eval\_2026-03-16\_23-34-44/  
eval.log — vaihingen / Adapter: mIoU=0.8152, OA=0.9048
- experiments/sam3\_linear\_probing\_vaihingen/  
sam3\_linear\_probing\_vaihingen\_eval\_2026-02-26\_13-57-59/eval.log — vaihingen / Linear  
Probing: mIoU=0.6917, OA=0.8186
- experiments/sam3\_lora\_vaihingen/sam3\_lora\_vaihingen\_eval\_2026-02-26\_13-47-26/eval.log —  
vaihingen / LoRA: mIoU=0.8326, OA=0.9101
- experiments/sam3\_adapter\_whu\_building/eval\_2026-03-24\_11-07-39/eval.log — whu\_building /  
Adapter: mIoU=0.9403, OA=0.9837
- experiments/sam3\_linear\_probing\_whu\_building/eval\_2026-03-12\_19-16-13/eval.log —  
whu\_building / Linear Probing: mIoU=0.8877, OA=0.9682
- experiments/sam3\_lora\_whu\_building/sam3\_lora\_whu\_building\_eval\_2026-03-11\_23-36-25/eval.log  
— whu\_building / LoRA: mIoU=0.9492, OA=0.9862